

Bihar Engineering University, Patna
B.Tech 3rd Semester Examination, 2025

Course: B.Tech
Code: 105303

Subject: Object Oriented Programming Using Java

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

- (a) Which of the following is not a core feature of Object-Oriented Programming?
 - (i) Inheritance
 - (ii) Encapsulation
 - (iii) Polymorphism
 - (iv) Compilation
- (b) What is the main purpose of a class in Java?
 - (i) To create objects
 - (ii) To store data only
 - (iii) To execute the program
 - (iv) To handle exceptions
- (c) Which access modifier provides the highest visibility in Java?
 - (i) Private
 - (ii) Protected
 - (iii) Public
 - (iv) Default
- (d) Method overloading is an example of
 - (i) Compile-time polymorphism
 - (ii) Runtime polymorphism
 - (iii) Inheritance
 - (iv) Abstraction
- (e) Method overriding demonstrates
 - (i) Compile-time polymorphism
 - (ii) Runtime polymorphism
 - (iii) Encapsulation
 - (iv) Abstraction
- (f) Which of the following is used to achieve 100% abstraction in Java?
 - (i) Abstract class
 - (ii) Interface
 - (iii) Final class
 - (iv) Static class
- (g) A class can implement
 - (i) Only one interface
 - (ii) Multiple interfaces
 - (iii) Only one abstract class
 - (iv) Multiple abstract classes
- (h) The Object class is
 - (i) The root of all classes in Java
 - (ii) A user-defined class
 - (iii) An interface
 - (iv) A final class only
- (i) Which keyword is used to import a package?
 - (i) package
 - (ii) import
 - (iii) include
 - (iv) using
- (j) String class in Java is
 - (i) Mutable
 - (ii) Immutable
 - (iii) Abstract
 - (iv) Final but mutable

Q.2 (a) Explain the different types of access modifiers available in Java and describe their visibility scopes with suitable examples.

[7]

(b) Explain the difference between procedural programming and object-oriented programming paradigms. Why is Java considered an object-oriented language?

[7]

- Q.3 (a) Describe the concept of method overloading and method overriding with the help of suitable examples from Java. [7]
(b) Explain the difference between an interface and an abstract class. How does Java support multiple inheritance using interfaces? [7]
- Q.4 (a) Describe the exception hierarchy in Java. Differentiate between checked and unchecked exceptions with examples. [7]
(b) Write a complete Java program to demonstrate the use of constructors (default, parameterized, copy), method overloading, and the 'this' keyword. [7]
- Q.5 (a) Write a Java program to handle multiple exceptions using nested try-catch blocks and the finally clause. Include re-throwing of exceptions. [7]
(b) Write a JDBC program to connect to a database, insert a record into a "Student" table, and retrieve all records using the Collections Framework (ArrayList) to store the results. [7]
- Q.6 (a) Compare and contrast inheritance hierarchies with interface implementation. Analyze when you would prefer interfaces over abstract classes in a Java project. [7]
(b) Evaluate the impact of garbage collection and the finalize() method on Java program performance. Is it advisable to rely on finalize() for resource cleanup? Justify your answer. [7]
- Q.7 (a) "Multithreading improves performance but introduces complexity." Critically evaluate this statement with reference to thread synchronization, priorities, and inter-thread communication. [7]
(b) Explain the key features of Object-Oriented Programming and describe how each feature provides advantages over procedural programming. [7]
- Q.8 (a) What is the purpose of break and continue statements in loops? Explain with examples. [7]
(b) Write a Java program to print all odd numbers between 1 to 50 using a while loop. [7]
- Q.9 Write short note on *any two* of the following: [7 x 2 = 14]
(a) History of java
(b) Concept of Packages in Java
(c) Polymorphism
(d) Multiple inheritance
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Bihar Engineering University, Patna
B.Tech 3rd Semester Examination, 2025

Course: B.Tech
Code: 105301

Subject: Digital Electronics

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven questions only)

[2 x 7 = 14]

- (a) A Programmable Logic Array consists of
- | | |
|--|--|
| (i) Fixed AND and fixed OR arrays | (iii) Fixed AND and programmable OR arrays |
| (ii) Programmable AND and programmable OR arrays | (iv) Programmable AND and fixed OR arrays |
- (b) The invalid condition in an SR flip-flop occurs when
- | | |
|---------------------|----------------------|
| (i) $S = 0, R = 0$ | (iii) $S = 1, R = 0$ |
| (ii) $S = 0, R = 1$ | (iv) $S = 1, R = 1$ |
- (c) A ripple counter is also known as
- | | |
|-------------------------|----------------------------|
| (i) Synchronous counter | (iii) Asynchronous counter |
| (ii) Up-down counter | (iv) Ring counter |
- (d) The resolution of an n-bit DAC is
- | | |
|--------------------|----------------|
| (i) 2^n | (iii) $1/2^n$ |
| (ii) $1/(2^n - 1)$ | (iv) 2^{n-1} |
- (e) Which memory is volatile in nature?
- | | |
|----------|------------|
| (i) ROM | (iii) PROM |
| (ii) RAM | (iv) EPROM |
- (f) Find the 2's complement of the binary number 1010
- | | |
|-----------|------------|
| (i) 0101 | (iii) 1110 |
| (ii) 0110 | (iv) 0111 |
- (g) Which logic family is most suitable for battery-operated portable devices, and why?
- | | |
|---------------------------|------------------------------------|
| (i) TTL – high speed | (iii) CMOS – low power consumption |
| (ii) RTL – low complexity | (iv) DTL – low cost |
- (h) A Multiplexer (MUX) is a device that
- | | |
|--|---|
| (i) Converts parallel data into serial | (iii) Selects one input from many and routes it to the output |
| (ii) Performs arithmetic operations | (iv) Stores data |
- (i) Evaluate the Boolean expression $Y = A + A \cdot B$
- | | |
|--------------|---------|
| (i) A | (iii) B |
| (ii) $A + B$ | (iv) 0 |
- (j) What is the output of a half adder for inputs $A = 1$ and $B = 1$?
- | | |
|-------------------------|--------------------------|
| (i) Sum = 0, Carry = 1 | (iii) Sum = 1, Carry = 1 |
| (ii) Sum = 1, Carry = 0 | (iv) Sum = 0, Carry = 0 |

- Q.2 (a) Implement the logic expression using a suitable Multiplexer $F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$. Using variables A, C, and D as a select line. [7]
- (b) Describe the architecture and working of Programmable Logic Array. [7]
- Q.3 (a) Design a synchronous counter using a T-flip-flop that counts $0 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 0$. [7]
- (b) Describe the dual slope ADC and explain why it is suitable for digital voltmeters. [7]
- Q.4 (a) Describe the CMOS logic family and its advantages over TTL. Discuss its role in modern VLSI design. [7]
- (b) Simplify the Boolean function: $F(A, B, C, D) = \sum(0, 2, 4, 5, 6)$. [7]
- Q.5 (a) Implement a full-adder circuit with a decoder and two OR gates. [7]
- (b) Compare PLA, PAL, and ROM-based implementations. [7]
- Q.6 (a) Design the logic expression $F = A(B + CD) + BC'$, using NAND gates only. [7]
- (b) Describe charge-coupled device memory and its working principle. [7]
- Q.7 (a) Explain the working of encoders and decoders. Design a 3-to-8 decoder logic circuit. [7]
- (b) Explain the concept of memory organization and operation in digital systems. [7]
- Q.8 (a) Explain the concept of the race-around condition in the JK flip-flop and suggest methods to eliminate it. [7]
- (b) Explain the R-2R ladder DAC and compare it with the weighted resistor DAC. [7]
- Q.9 Write short notes on **any two** of the following: [7 x 2 = 14]
- Flip flops
 - sample-and-hold circuit
 - PROM, EPROM and EEPROM
 - BCD to Gray code conversion